Koya Sakamoto

Ph.D. Student in Computer Science

The University of Tokyo · Embodied AI, Active Perception, Robot Manipulation

kohya0717@gmail.com
k0uya.github.io
koya-sakamoto-943871214
Tokyo, Japan

RESEARCH INTERESTS

Active perception for embodied agents, vision-language models for robotics, robot manipulation, and 3D scene understanding.

EDUCATION

The University of Tokyo — Apr 2025 – Present
Ph.D. in Engineering (Computer Science), Department of Technology Management for Innovation — *Tokyo, Japan*

- Advisor: Prof. Yusuke Iwasawa. Expected graduation: Mar 2028.

Kyoto University — Apr 2023 – Mar 2025
M.Eng. in Informatics, Graduate School of Informatics — *Kyoto, Japan*

- Advisor: Prof. Shin Ishii. Collaborative graduate student at ATR (Prof. Motoaki Kawanabe).

Kyoto University — Apr 2019 – Mar 2023
B.Eng. in Informatics and Mathematical Science, Faculty of Engineering — *Kyoto, Japan*

- Thesis advisor: Prof. Hidetoshi Shimodaira.

RESEARCH EXPERIENCE

Stanford University — Nov 2026 – Dec 2026 (upcoming)
Visiting Student Researcher, Geometric Computation Group (Prof. Leonidas Guibas) — *Stanford, CA, USA*

- Two-month research visit hosted by Prof. Leonidas Guibas, in collaboration with Yining Hong.

The University of Tokyo — Apr 2025 – Present
Ph.D. research, Matsuo–Iwasawa Laboratory (Prof. Yusuke Iwasawa) — *Tokyo, Japan*

Rendering-Free Lookahead (RFL) for question-guided active vision — May 2026 – Present

- Proposed a viewpoint-selection policy that ranks candidate camera motions by their predicted future answerability, distilled from a privileged teacher that renders future views in 3DGS scenes.
- Improved the mean judge score by 43% over a direct-action baseline on E3VS-Bench; under review [S1].

E3VS-Bench: viewpoint-dependent active perception in 3D Gaussian Splatting scenes — Nov 2024 – Mar 2026

- Built a benchmark in which a VLM must control a 5-DoF camera inside 3D Gaussian Splatting scenes to find the viewpoints that reveal the answer to a question.
- First-author paper accepted at ECCV 2026 [C1].

Advanced Telecommunications Research Institute International (ATR) — Jun 2022 – Oct 2024
Department of Dynamic Brain Imaging (Prof. Motoaki Kawanabe) — *Kyoto, Japan*

Map-based modular approach for zero-shot embodied question answering — Jun 2022 – Oct 2024

- Proposed a modular framework that combines a semantic map with pretrained models to answer questions about a 3D environment without task-specific training.
- Validated the framework both in simulation and on a real robot; first-author paper at IROS 2024 [C4].

Kyoto University — Apr 2022 – Mar 2023
Bachelor's thesis, Shimodaira Laboratory (Prof. Hidetoshi Shimodaira) — *Kyoto, Japan*

Decomposing NeRF with view-dependent queries — Apr 2022 – Mar 2023

- Built a pipeline that decomposes a NeRF scene into objects by applying a referring-expression comprehension model to rendered views.

WORK EXPERIENCE

OMRON SINIC X Corporation (OSX) — Apr 2026 – Present
Research Intern — *Tokyo, Japan*

- Research on active vision for robotic manipulation.

AI Robot Association (AIRoA) Apr 2025 – Sep 2025
Research Assistant *Tokyo, Japan*

- Built a data pipeline that converts rosbag recordings into the LeRobot format for robot-policy training.
- Contributed to the AIRoA MoMa dataset, a large-scale hierarchical dataset for mobile manipulation [P1].

Matsuo Institute, Inc. Apr 2023 – Mar 2025
Research Intern *Tokyo, Japan*

- Joint research with TIER IV, Inc. on neural 3D reconstruction for autonomous driving.
- Developed and evaluated a neural simulator with traffic agents for autonomous-driving systems [D1, D2].

Advanced Telecommunications Research Institute International (ATR) Jun 2022 – Mar 2023
Research Intern *Kyoto, Japan*

- Implemented and deployed an embodied question answering (EQA) system on a real robot.

PUBLICATIONS

International conferences (peer-reviewed)

[C1] **E3VS-Bench: A Benchmark for Viewpoint-Dependent Active Perception in 3D Gaussian Splatting Scenes**
Koya Sakamoto, Taiki Miyanishi, Daichi Azuma, Shuhei Kurita, Shu Morikuni, Naoya Chiba, Motoaki Kawanabe, Yusuke Iwasawa, Yutaka Matsuo
European Conference on Computer Vision (ECCV), 2026

[C2] **CityNav: Language-Goal Aerial Navigation Dataset with Geographic Information**
Jungdae Lee, Taiki Miyanishi, Shuhei Kurita, **Koya Sakamoto**, Daichi Azuma, Yutaka Matsuo, Nakamasa Inoue
IEEE/CVF International Conference on Computer Vision (ICCV), 2025

[C3] **GeoProg3D: Compositional Visual Reasoning for City-Scale 3D Language Fields**
Shunsuke Yasuki, Taiki Miyanishi, Nakamasa Inoue, Shuhei Kurita, **Koya Sakamoto**, Daichi Azuma, Masato Taki, Yutaka Matsuo
IEEE/CVF International Conference on Computer Vision (ICCV), 2025

[C4] **Map-based Modular Approach for Zero-shot Embodied Question Answering**
Koya Sakamoto, Daichi Azuma, Taiki Miyanishi, Shuhei Kurita, Motoaki Kawanabe
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2024

[C5] **Answerability Fields: Answerable Location Estimation via Diffusion Models**
Daichi Azuma, Taiki Miyanishi, Shuhei Kurita, **Koya Sakamoto**, Motoaki Kawanabe
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2024

Under review

[S1] **Rendering-Free Lookahead for Question-Guided Active Vision**
Koya Sakamoto, Daichi Azuma, Shuhei Kurita, Naoya Chiba, Yusuke Iwasawa, Yutaka Matsuo, Taiki Miyanishi
Under review · arXiv:2610.11039

Preprints

[P1] **NavWAM: A Navigation World Action Model for Goal-Conditioned Visual Navigation**
Daichi Azuma, Taiki Miyanishi, **Koya Sakamoto**, Shuhei Kurita, Yaonan Zhu, Petr Khrapchenkov, Motoaki Kawanabe, Yusuke Iwasawa, Yutaka Matsuo
arXiv preprint arXiv:2606.13494, 2026

[P2] **AIRoA MoMa Dataset: A Large-Scale Hierarchical Dataset for Mobile Manipulation**
Ryosuke Takanami, Petr Khrapchenkov, Shu Morikuni, Jumpei Arima, …, **Koya Sakamoto**, …, Tatsuya Matsushima, Yusuke Iwasawa, Yutaka Matsuo, Tetsuya Ogata (35 authors)
arXiv preprint arXiv:2509.25032, 2025

Domestic conferences (Japan)

[D1] **Building and Evaluating a Neural Simulator with Traffic Agents for Autonomous Driving**
Koya Sakamoto, Mae Yamaguchi, Yoshiki Ito, Akihito Ohsato, Shintaro Tomie, Shintaro Sakoda, Masaya Kataoka, Makoto Kawano
The 28th Meeting on Image Recognition and Understanding (MIRU), 2025

[D2] **Development and Evaluation of Neural Simulator for Autonomous Driving Systems**
Junnosuke Kamohara, **Koya Sakamoto**, Akihito Ohsato, Shintaro Tomie, Masaya Kataoka, Makoto Kawano
The 38th Annual Conference of the Japanese Society for Artificial Intelligence (JSAI), 2024

SKILLS

Robotics & Hardware	TurtleBot3, xArm6, reBot Arm B601-DM, SO-101, Intel RealSense, DJI Tello, 360-degree camera
Languages	Japanese (native), English (TOEFL iBT 89, Nov 2024; 92, Oct 2022)